MuSc: Zero-Shot Industrial Anomaly Classification and Segmentation with Mutual Scoring of the Unlabeled Images

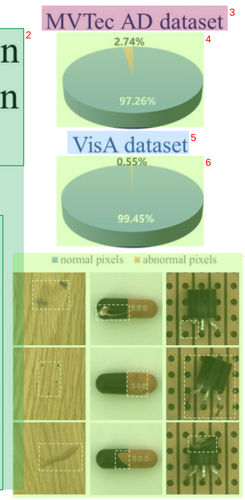
Xurui Li[1,*], Ziming Huang[1,*], Feng Xue[3], Yu Zhou[1,2]

[1]Huazhong University of Science and Technology, [2]Artificial Intelligence Research Institute, Wuhan JingCe Electronic Group Co.,LTD, [3]University of Trento

{xrli_plus, zmhuang, yuzhou}@hust.edu.cn}, feng.xue@unitn.it

Motivation

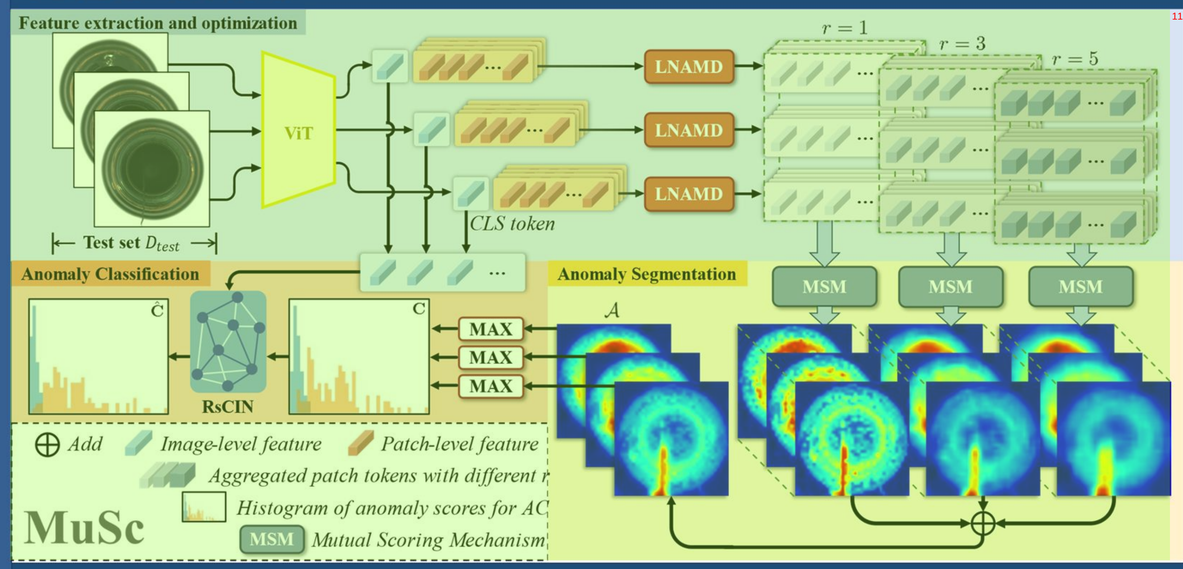
- A **rich amount of** normal information implicit in unlabeled test images can be exploited for anomaly detection.
- **A discriminative characteristic**: the normal image patches could find a relatively large number of similar patches in other unlabeled images, while the abnormal ones only have a few similar patches.



Contributions

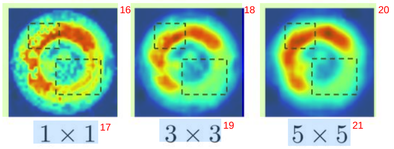
- **The first method** that **only uses the unlabeled test images** for industrial anomaly detection.
- We reveal the potential capability of normal and abnormal patches contained in unlabeled images.
- **21.1% PRO gains** and **21.9% seg-AP gains** on MVTec AD and **19.4% seg-AP gains** on VisA.

Framework



LNAMD

- Neighborhood range
  $1\times1 \quad 3\times3 \quad 5\times5$
- Aggregate function:
  Adaptive average pooling
- Multi-degree aggregation
  To segment abnormal regions of varying sizes.



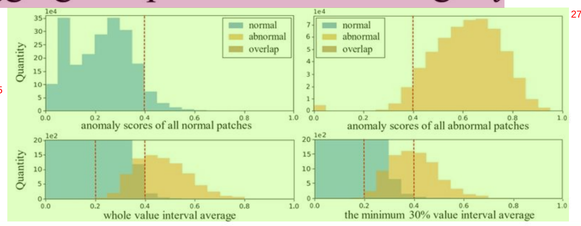
Mutual Scoring Mechanism

- Image I_j assigns a score to each aggregated patch token of image I_i

$$a_{i,l}^{m,r}(I_j) = \min_n \|\hat{p}_{i,l}^{m,r} - \hat{p}_{j,l}^{n,r}\|_2$$

- Interval Average (minimum 30%)

$$\bar{a}_{i,l}^{m,r} = \frac{1}{K}\sum_{k\in[1,K]} a_{i,l}^{m,r}(\bar{I}_k)$$



- Mean:
  r and l

$$a_i^m = \frac{1}{L}\sum_{l\in\{1,\dots,L\}}\frac{1}{3}\sum_{r\in\{1,3,5\}}\bar{a}_{i,l}^{m,r}$$

Experiments

- Comparison with 0-shot methods

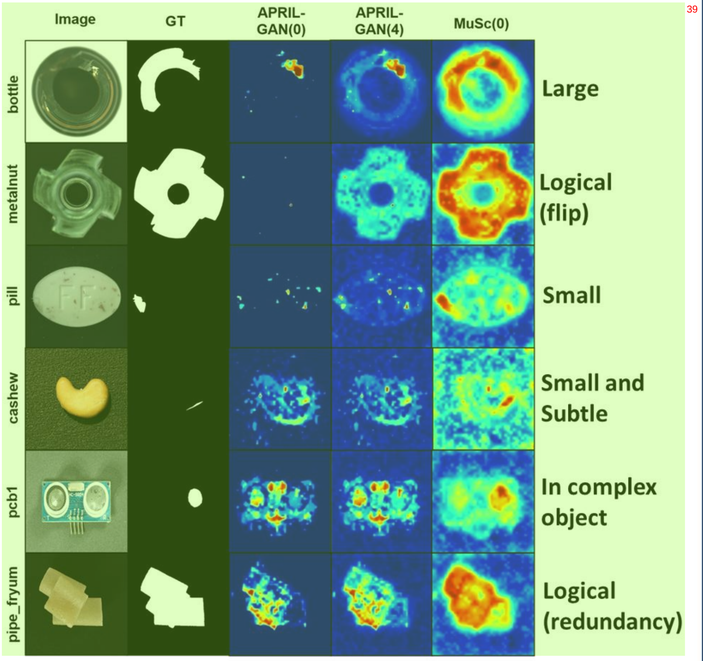
Methods	WinCLIP[1]	APRIL-GAN[2]	ACR[3]	MuSc(ours)
MVTec AD Dataset				
cls-AUROC	91.8	86.1	85.8	**97.8(+6.0)**
PRO	64.6	44.0	72.7	**93.8(+12.4)**
VisA Dataset				
cls-AUROC	78.1	78.0	-	**92.8(+10.7)**
PRO	56.8	86.8	-	**92.7(+5.7)**

No training **No prompt**

- Comparison with 4-shot methods

Methods	PatchCore[4]	WinCLIP[1]	APRIL-GAN[2]	MuSc(ours)
MVTec AD Dataset				
cls-AUROC	88.8	95.2	92.8	**97.8(+2.6)**
PRO	84.3	89.0	91.8	**93.8(+2.0)**
VisA Dataset				
cls-AUROC	85.3	87.3	92.6	**92.8(+0.2)**
PRO	84.9	87.6	90.2	**92.7(+2.5)**

Detect **various types of defects**



RsCIN

- Image-level feature $\mathcal{F}_i$
  Extracted by ViT
- Multi-window Mask

$$M_k(i,j) = \begin{cases} 1, & \text{if } I_j \in \mathcal{N}_k(I_i) \\ 0, & \text{otherwise,} \end{cases} \qquad k \in \{k_1, k_2\}$$

- Re-Scoring

$$D(i,i) = \sum_{j=1}^{N} M_k \odot W(i,j)$$

$$\hat{\mathbf{C}} = \Big(\sum_{M_k\in\overline{M}} (D^{-1}(M_k\odot W)\mathbf{C}) + \mathbf{C}\Big)/(K+1)$$

- Similarity matrix

$$W_{i,j} = \mathcal{F}_i \cdot \mathcal{F}_j$$

Optimize score c_i of I_i

Optimized score $\hat{c}_i$

$$\hat{c}_i = \frac{c_i}{3} + \frac{1}{3}\sum_{j=1}^{k_2}\overline{w}_{i,j}\bar{c}_j$$

Weighted average of scores within the window mask

$$\overline{w}_{i,j} = \begin{cases} \hat{w}_{i,j}^{k_1} + \hat{w}_{i,j}^{k_2}, & \text{if } 0 < j \le k_1 \\ \hat{w}_{i,j}^{k_2}, & \text{if } k_1 < j \le k_2 \end{cases}$$

Universal module for other AD methods

References

[1] Jeong J, Zou Y, Kim T, et al. Winclip: Zero-/few-shot anomaly classification and segmentation[C]//Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition. 2023: 19606-19616.

[2] Chen X, Han Y, Zhang J. A zero-/few-shot anomaly classification and segmentation method for cvpr 2023 vand workshop challenge tracks 1&2: 1st place on zero-shot ad and 4th place on few-shot ad[J]. arXiv preprint arXiv:2305.17382, 2023.

[3] Li A, Qiu C, Kloft M, et al. Zero-shot anomaly detection via batch normalization[J]. Advances in Neural Information Processing Systems, 2024, 36.

[4] Roth K, Pemula L, Zepeda J, et al. Towards total recall in industrial anomaly detection[C]//Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition. 2022: 14318-14328.

Paper

Code